CSE2331 Final Exam Review 2

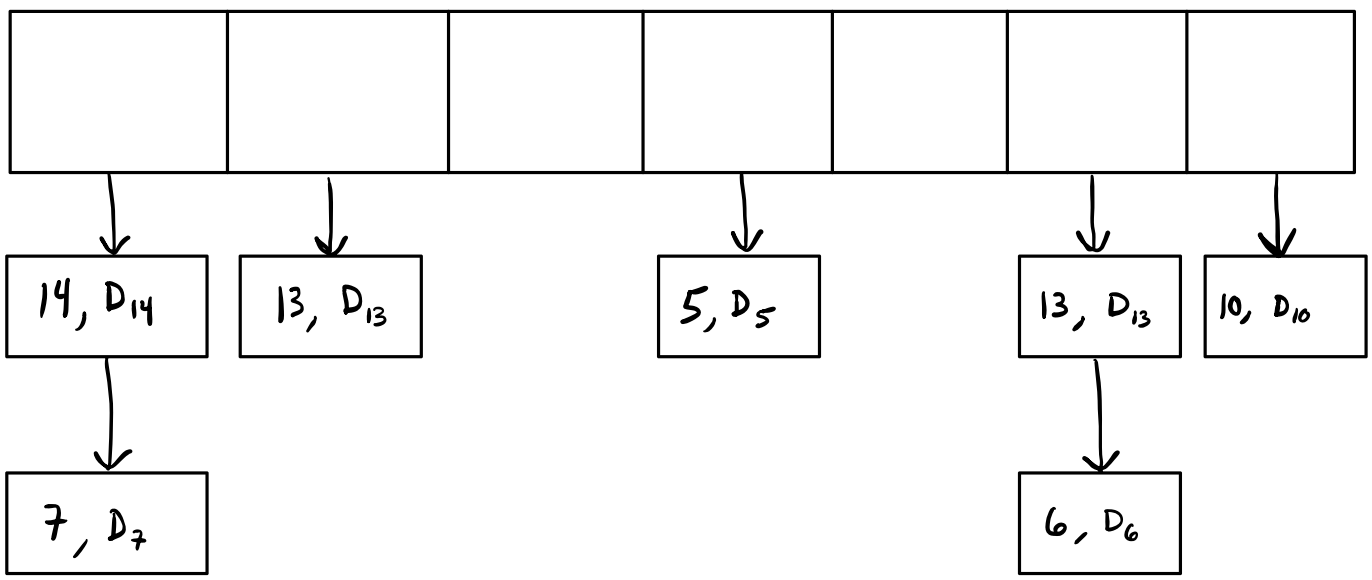
Problem 1. Let H be an initially empty hash table of size 7 implemented using chained hashing. Assume that the hash function for the hash table is

$$h(K) = 2K \bmod 7$$

Perform the following sequence of operations in order.

1: `HashTable.Insert`$(5, D_5)$ $h(5) = 10 \bmod 7 = 3$
2: `HashTable.Insert`$(10, D_{10})$ $h(10) = 20 \bmod 7 = 6$
3: `HashTable.Insert`$(11, D_{11})$ $h(11) = 22 \bmod 7 = 1$
4: `HashTable.Insert`$(13, D_{13})$ $h(13) = 26 \bmod 7 = 5$
5: `HashTable.Insert`$(6, D_6)$ $h(6) = 12 \bmod 7 = 5$
6: `HashTable.Insert`$(14, D_{14})$ $h(14) = 28 \bmod 7 = 0$
7: `HashTable.Insert`$(7, D_7)$ $h(7) = 14 \bmod 7 = 0$

and draw the resulting hash table.



Problem 2. Let H be an initially empty hash table of size 7 implemented using open address hashing with hash function

$$h(K, j) = (2K + 3j) \bmod 7.$$

Perform the following sequence of operations in order on the Hash Table

1: `HashTable.Insert`$(5, D_5)$
2: `HashTable.Insert`$(10, D_{10})$
3: `HashTable.Insert`$(11, D_{11})$
4: `HashTable.Insert`$(13, D_{13})$
5: `HashTable.Insert`$(6, D_6)$
6: `HashTable.Insert`$(14, D_{14})$
7: `HashTable.Insert`$(7, D_7)$

and draw the resulting hash table.

$14, D_{14}$	$11, D_{11}$	$7, D_7$	$5, D_5$	$6, D_6$	$13, D_{13}$	$10, D_{10}$

$h(5,0) = 10 \bmod 7 = 3$ ✓
$h(10,0) = 20 \bmod 7 = 6$ ✓
$h(11,0) = 22 \bmod 7 = 1$ ✓

$h(13,0) = 26 \bmod 7 = 5$ ✓

$h(6,0) = 12 \bmod 7 = 5$ ✗
$h(6,1) = 15 \bmod 7 = 1$ ✗
$h(6,2) = 18 \bmod 7 = 4$ ✓

$h(14,0) = 28 \bmod 7 = 0$ ✓

$h(7,0) = 14 \bmod 7 = 0$ ✗
$h(7,1) = 17 \bmod 7 = 3$ ✗
$h(7,2) = 20 \bmod 7 = 6$ ✗
$h(7,3) = 23 \bmod 7 = 2$ ✓

Problem 3. Suppose H is Hash Table of size m implemented using Chained Hashing Why is the hash function

$$h(K) = \left(\sum_{i=1}^{K} i\right) \bmod m$$

a poor choice hash function? What modification could be made to this hash function to make it a good hash function?

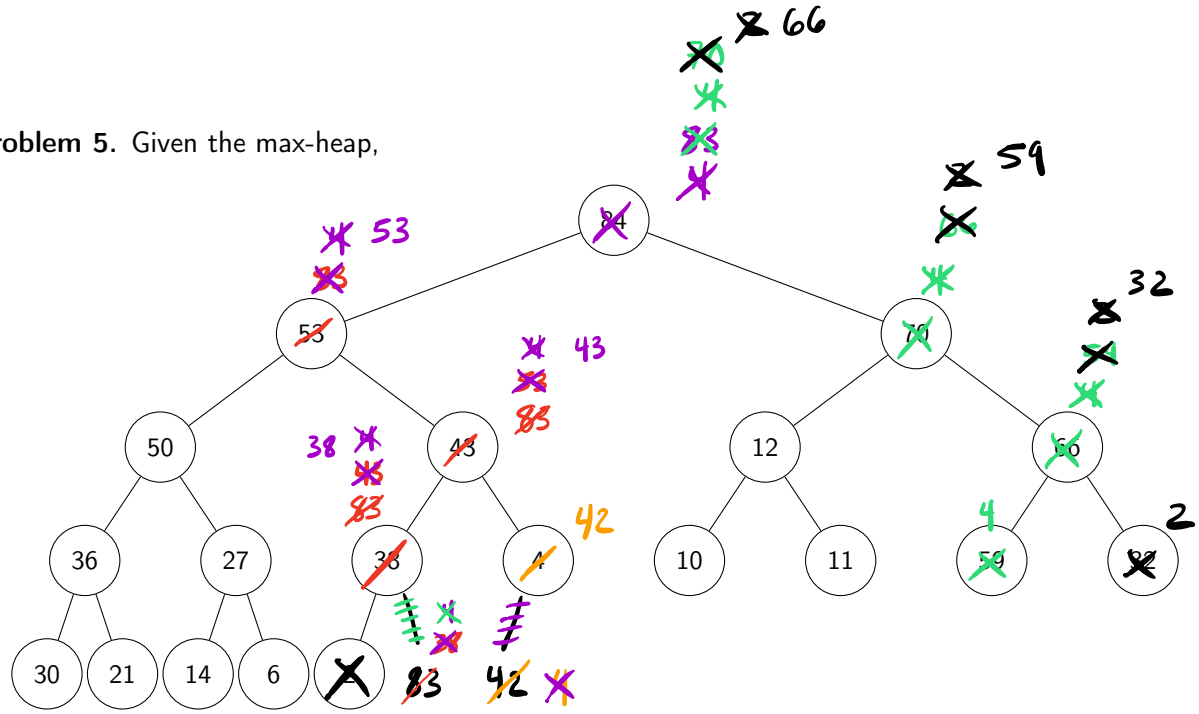
takes $\Theta(K)$ time to compute

rewrite as $h(K) = \frac{K(K+1)}{2} \bmod m$

Problem 4. What is one reason to use a Max-heap instead of a hash table?

Heap sort!

Problem 5. Given the max-heap,

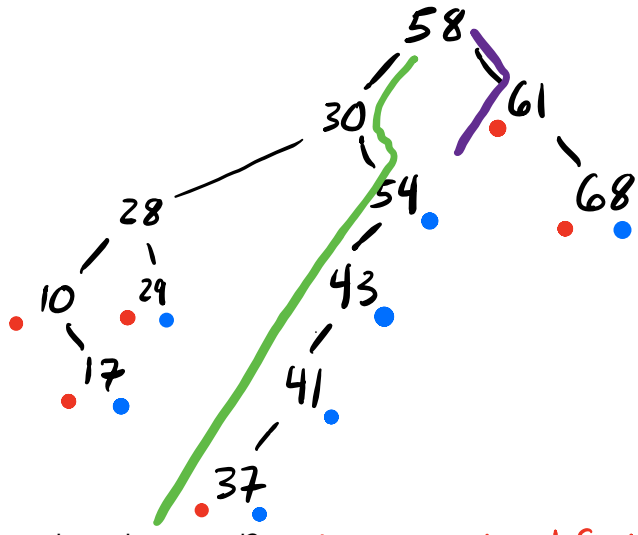


Perform the following sequence of operations of the max-heap:

1: `Heap.Insert`(83)
2: `Heap.Insert`(42)
3: `Heap.ExtractMax()`
4: `Heap.ExtractMax()`
5: `Heap.ExtractMax()`

Problem 6. Let T be an initially empty binary search tree. Perform the following sequence of operations in order.

```
BST.Insert(58)
BST.Insert(30)
BST.Insert(28)
BST.Insert(54)
BST.Insert(43)
BST.Insert(41)
BST.Insert(61)
BST.Insert(29)
BST.Insert(68)
BST.Insert(10)
BST.Insert(37)
BST.Insert(17)
```



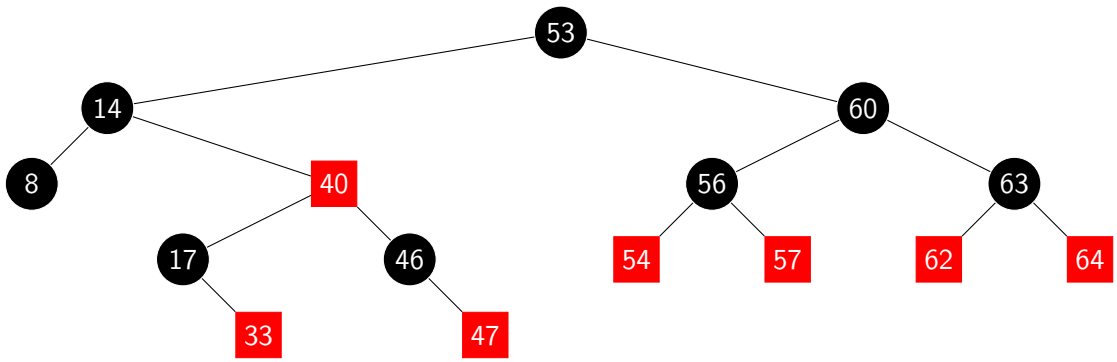
Which nodes in the tree cannot be right rotated? • All nodes w/ no left child.

Which nodes in the tree cannot be left rotated? ● All nodes w/ no right child.

Can you color the tree as a **red-black** tree? If yes, color it. If not, explain why not.

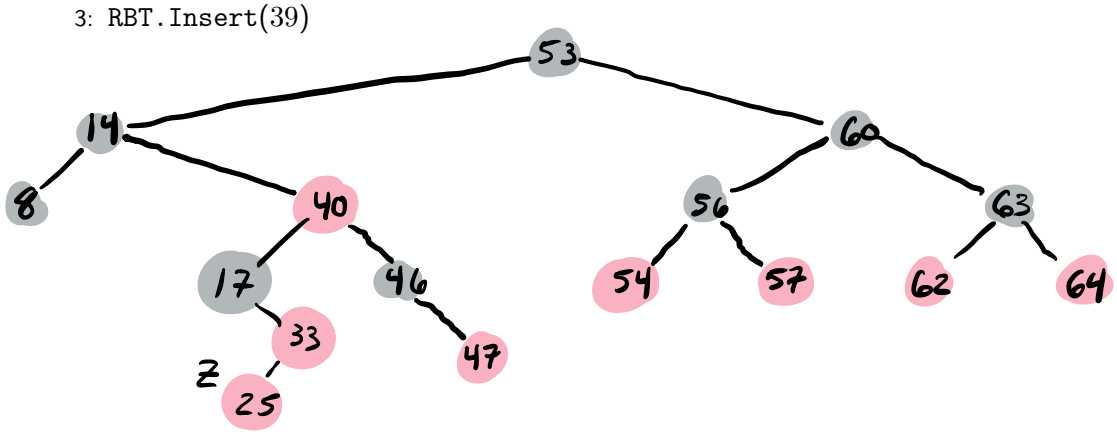
Nope! The path 58 → 61 → NIL has at most 3 black nodes, but the path 58 → 30 → 54 → 43 → 41 → 37 → NIL has at least 4.

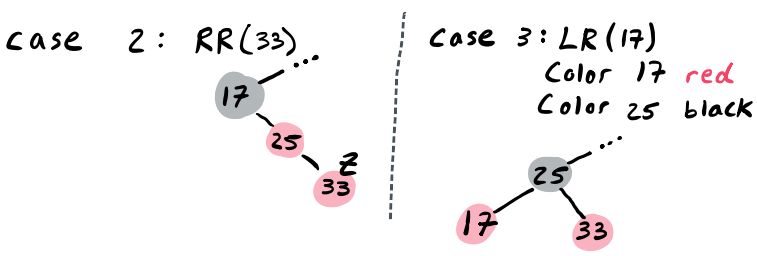
Problem 7. Given the red-black tree,

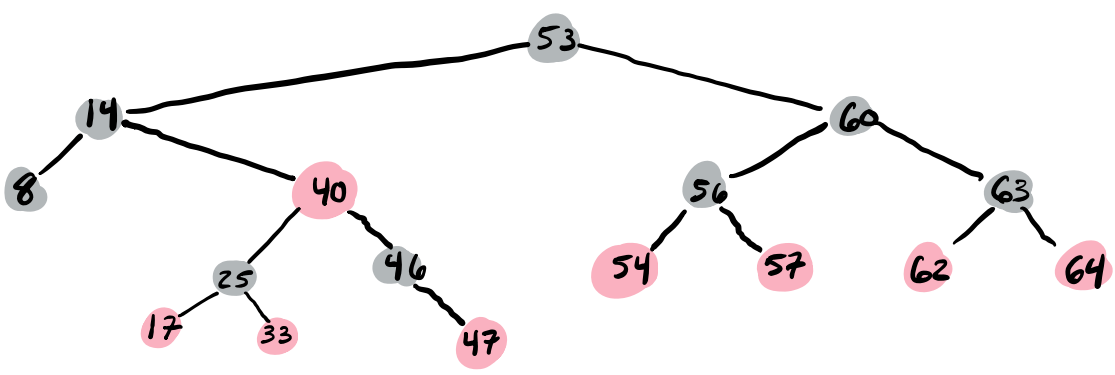


Perform the following sequence of operations in order.

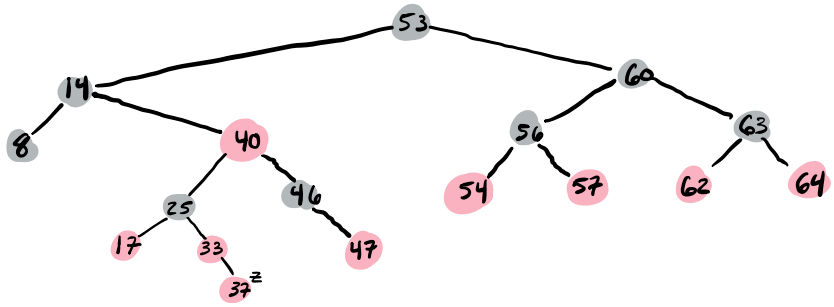
1: `RBT.Insert`(25)
2: `RBT.Insert`(37)
3: `RBT.Insert`(39)





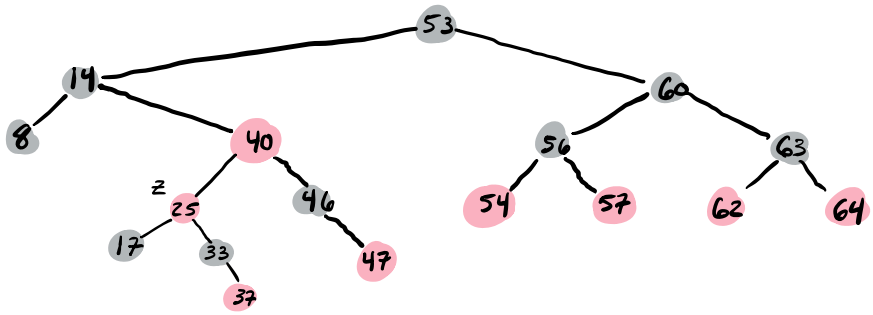


Insert (37)

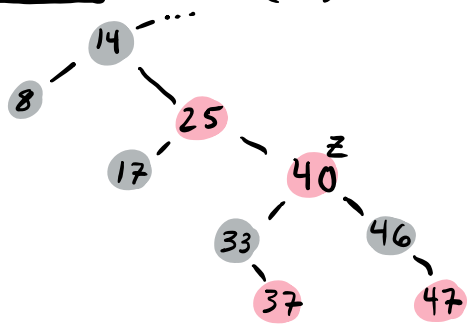


Case 1: Color 17, 33 red

Color 25 black



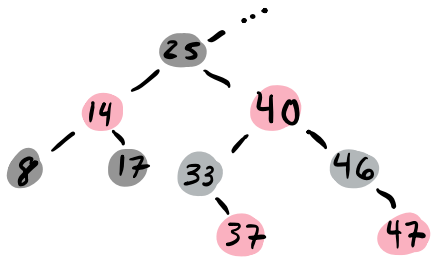
Case 2 : RR (40)

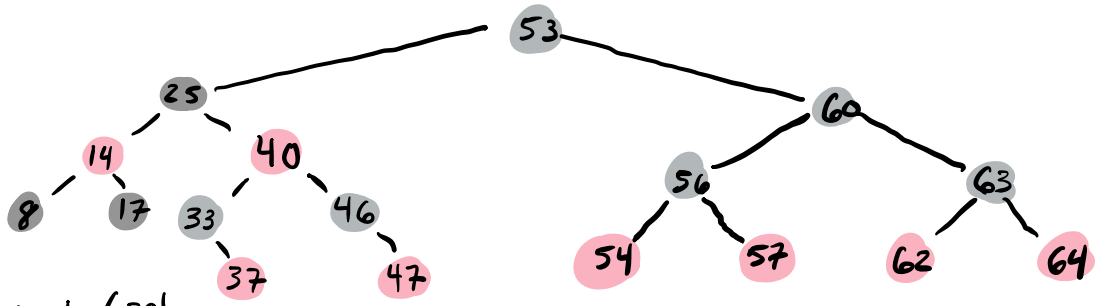


Case 3 : LR(14)

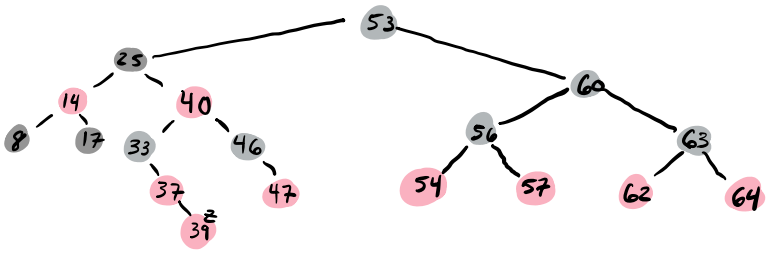
Color 14 red

Color 25 black





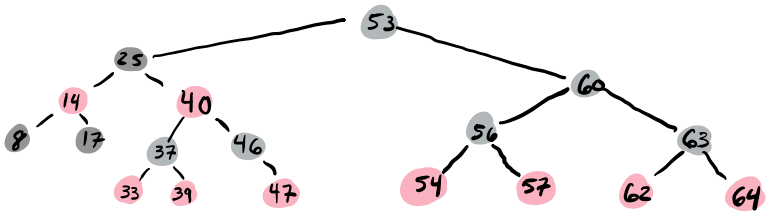
Insert (39)



Case 3 : LR(33)

Color 33 red

Color 37 black



Problem 8. Consider the algorithm given below

```
function Func(A[],n,K)
    HashTable.Init()
    for i = 1 to n do
        if HashTable.Member(A[i]) then
            Print "Duplicate value found, not inserting."
        else
            HashTable.Insert(A[i],true)
        end if
    end for
    for i = 1 to n do
        if HashTable.Member(K − A[i]) then
            return true
        end if
    end for
    return false
end function
```

$s = 1$

$s = n$

$s = n$

Assume the Hash Table is implemented using Open Address Hashing and that the size of the Hash Table is at least $2n$. Determine the best, expected, and worst case running time.

BC: All methods take $\Theta(1)$ time + return immediately

$B(n) = \sum_{i=1}^{n} c = cn$ Thus, $B(n) \in \Theta(n^2)$.

EC: All methods take $\Theta(1)$ time. 2nd loop runs at most n times

$ET(n) \leq \sum_{i=1}^{n} c + \sum_{i=1}^{n} c = 2cn \Big\} O(n)$

BC is a LB so $ET(n) \in \Omega(n)$

Thus, $ET(n) \in \Theta(n)$

WC: All methods take $\Theta(s)$ time.

$W(n) = \sum_{s=1}^{n} cs + \sum_{i=1}^{n} cn = c\frac{n(n+1)}{2} + cn^2$

Thus, $W(n) \in \Theta(n^2)$.

Problem 9. Consider the following problem:

```
function Func(A[],n)
    P.Init()
    i = 1
    while i ≤ n do
        for j = 1 to n do
            P.Insert(A[i] · A[j])
        end for
        i = i + 5
    end while
    for i = 1 to n do
        x = P.ExtractMax()
        Print x
    end for
end function
```

$s = 1$

$s = n \cdot \frac{n}{5} = \frac{n^2}{5}$

$s = \frac{n^2}{5} - n$

Iteration	i
0	1
1	6
2	11
$\vdots$	$\vdots$
γ	$1+5\gamma$

Stops when

$1 + 5\gamma = n$

$\gamma = \frac{n-1}{5} \approx \frac{n}{5}$

Assume that s is the number of elements currently in the Priority Queue. Analyze the running time of the given algorithm assuming that `P.Insert(x)` takes $\Theta(\log(s))$ time and `P.ExtractMax()` takes $\Theta(s)$ time.

$$T(n) = \sum_{s=1}^{n^2/5} c\log(s) + \sum_{s=\frac{n^2}{5}-n}^{n^2/5} cs$$

let $x = \frac{n^2}{5}$

$$\leq \sum_{s=1}^{x} c\log(x) + \sum_{s=x-n}^{x} cx$$

$$\geq \sum_{s=x-n}^{x} c(x-n)$$

$$\leq x\log(x) + (n+1)cx$$

$$\geq (n+1)c(x-n)$$

$$\leq \frac{n^2}{5} c\log(n^2/5) + (n+1)cn^2/5$$

$$\geq (n+1)c(n^2/2 - n)$$

Thus, $T(n) \in \Theta(n^3)$.

Problem 10. Consider the following problem:

```
function Func(A[],i,j)
    P.Init()
    n = j − i + 1
    for k = i to j do
        P.Insert(A[k])
    end for
    while P.Size() > n/2 do
        x = P.ExtractMax()
        Print x
    end while
end function
```

$s = 1$

runs $j - i + 1 = n$ times

$s = n$

$s = n/2$

Assume that s is the number of elements currently in the Priority Queue. Analyze the running time of the given algorithm assuming that `P.Insert(x)` takes $\Theta(s)$ time and `P.ExtractMax()` takes $\Theta(s^2)$ time.

$$T(n) = \sum_{s=1}^{n} cs + \sum_{s=\frac{n}{2}}^{n} cs^2$$

$$\leq \sum_{s=1}^{n} cn + \sum_{s=\frac{n}{2}}^{n} cn^2 \qquad \geq \sum_{s=\frac{n}{2}}^{n} c\left(n/2\right)^2$$

$$\leq cn^2 + \left(\frac{n}{2}+1\right)cn^2 \qquad \geq \frac{n}{2} \cdot \frac{cn^2}{4}$$

Thus, $T(n) \in \Theta(n^3)$.

Problem 11. Explain as much as you can about ALL of the data structures we discussed in class.

Left for you all!